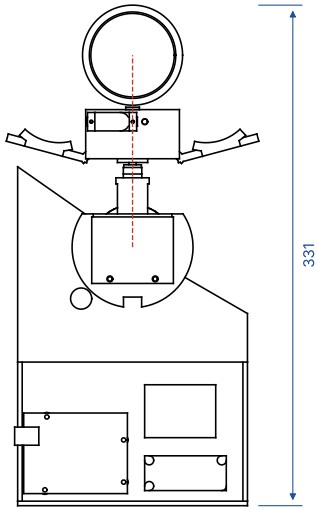
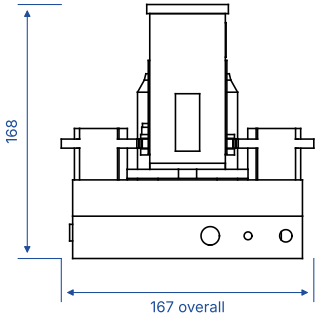
# CA-000  Concept A assembly, HOME pose (arm at START)

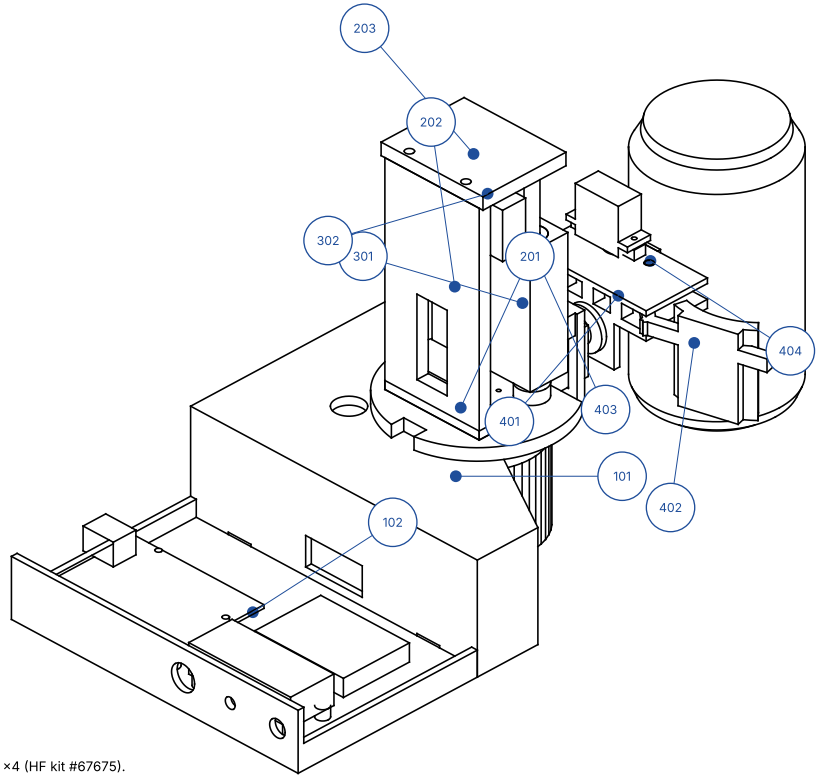
TOP VIEW




R 127 mm (5.0 in): turntable axis → can centre (dashed)




FRONT VIEW




ISOMETRIC (not to scale)

**PARTS LIST (full BOM: docs/05_bom.md)**

| | | | |
|---|---|---|---|
| CA-101 | Base | PETG | 1 |
| CA-102 | Electronics tray | PETG | 1 |
| CA-201 | Turret plate | PETG | 1 |
| CA-202 | Spine | PETG | 1 |
| CA-203 | Top cap | PETG | 1 |
| CA-301 | Carriage + arm | PETG | 1 |
| CA-302 | Lift link | PETG | 1 |
| CA-401 | Gripper frame | PETG | 1 |
| CA-402 | Finger, right | PETG | 1 |
| CA-403 | Finger, left | PETG | 1 |
| CA-404 | Can bumper | silicone | 1 |
| CA-P02 | Servo MG996R (turntable) | buy | 1 |
| CA-P03 | Servo MG90S (lift, wrist) | buy | 2 |
| CA-P04 | Servo SG90 (gripper) | buy | 1 |
| CA-P05 | Bearing 6807-2RS | buy | 1 |
| CA-P06 | Linear bearing LM6UU | buy | 4 |
| CA-P07 | Steel rod 6 × 112 mm | buy | 2 |
| CA-P01 | Arduino Uno R3 | buy | 1 |
| CA-P09 | Buck converter XL4015 | buy | 1 |

Balloon numbers = part number without the CA- prefix. Fasteners: #6-32 × ½″ ×8, #8-32 × ½″ ×4 (HF kit #67675).
Motion ranges: turret 0 → −90°, lift 0 → 15.3 mm, wrist 0 → 180°, fingers 0 → 75°.

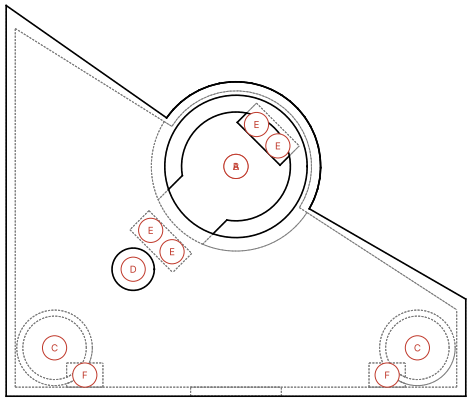
| | |
|---|---|
| PART NO. | **CA-000** |
| TITLE | **Concept A assembly (HOME pose)** |
| MASS | ≈ 571 g robot + 15 g can (CAD) |
| SCALE | 1:5 (front, top) |
| UNITS | mm |
| SOURCE | cad/out/concept_a_detail_assembly.step |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 1 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |

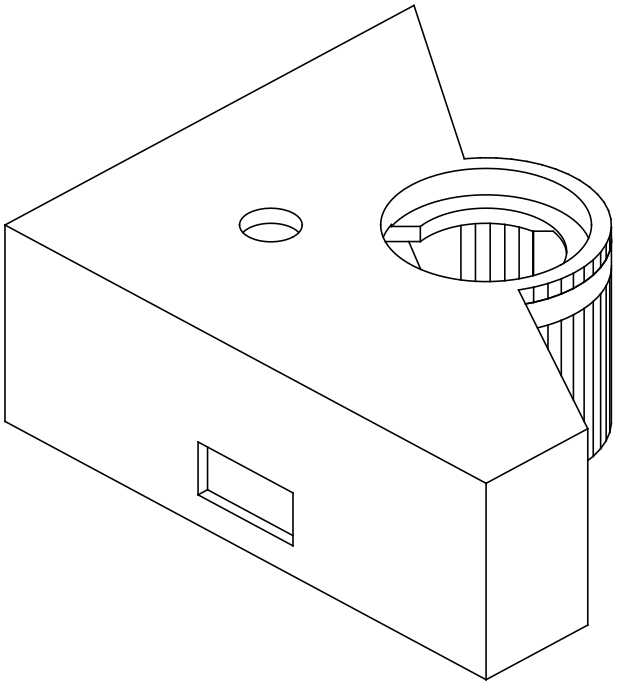
# CA-101 Base

**NOTES**

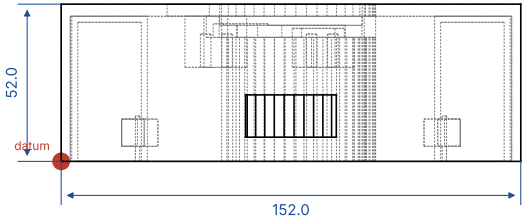
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

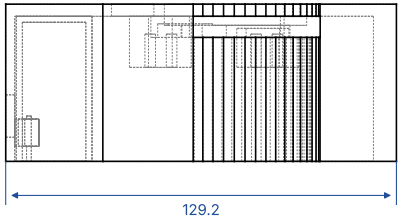


ISOMETRIC (not to scale)




FRONT VIEW




RIGHT VIEW

**HOLE TABLE (letters mark hole centres on the view along that axis)**

| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 47.10 | 1 | Z | 7.0 | (76.0, 76.0) | 6807 outer race press fit |
| B | 36.00 | 1 | Z | 4.0 | (76.0, 76.0) | bearing seat floor bore |
| C | 21.00 | 2 | Z | 46.0 | (16.0, 16.0); (136.0, 16.0) | ballast washer pocket |
| D | 14.00 | 1 | Z | 4.0 | (42.0, 42.0) | cable hole from the turret |
| E | 3.50 | 4 | Z | 11.0 | (47.8, 54.9); (54.9, 47.8) +2 | #8-32 pilot |
| F | 2.90 | 2 | Z | 15.0 | (26.0, 7.0); (126.0, 7.0) | #6-32 pilot (self-threading) |

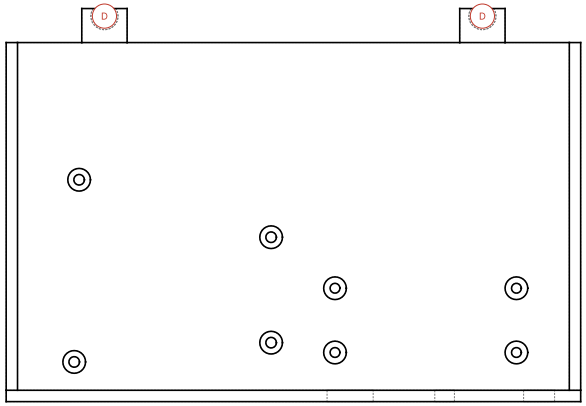
| | |
|---|---|
| PART NO. | **CA-101** |
| TITLE | **Base** |
| MATERIAL | PETG printed, 0.4 mm nozzle |
| MASS | 109.2 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 2:5 |
| UNITS | mm · general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/base.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 2 of 13 |
| DRAWN | ME 2331 Project 2 team 2026-11-02 |

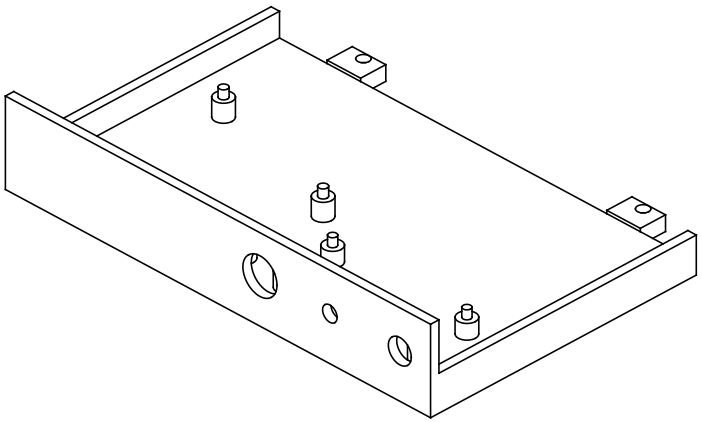
# CA-102  Electronics tray

**NOTES**

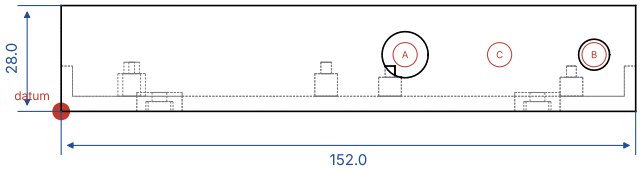
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

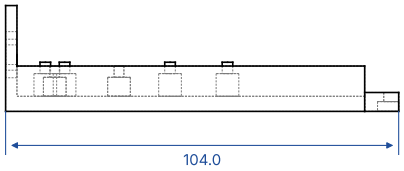


ISOMETRIC (not to scale)




FRONT VIEW




RIGHT VIEW

**HOLE TABLE  (letters mark hole centres on the view along that axis)**

| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 12.20 | 1 | Y | 3.0 | (91.0, 15.0) | 12 mm START button |
| B | 8.20 | 1 | Y | 3.0 | (141.0, 15.0) | DC panel jack |
| C | 5.20 | 1 | Y | 3.0 | (116.0, 15.0) | 5 mm status LED |
| D | 3.90 | 2 | Z | 2.4 | (26.0, 102.0); (126.0, 102.0) | #6-32 clearance / running |

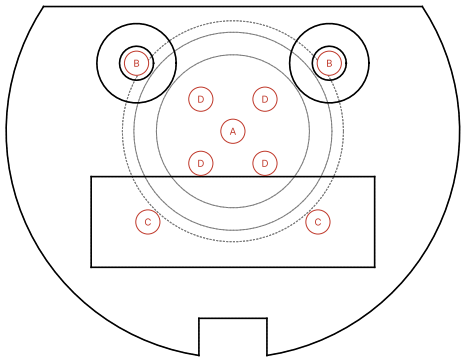
| | |
|---|---|
| PART NO. | **CA-102** |
| TITLE | **Electronics tray** |
| MATERIAL | PETG  printed, 0.4 mm nozzle |
| MASS | 57.0 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 1:2 |
| UNITS | mm  ·  general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/electronics_tray.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 3 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |

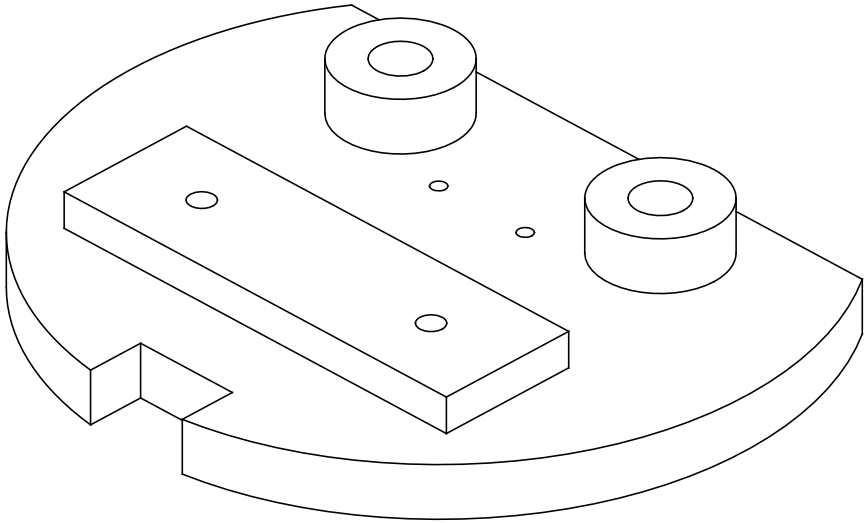
# CA-201 Turret plate

**NOTES**

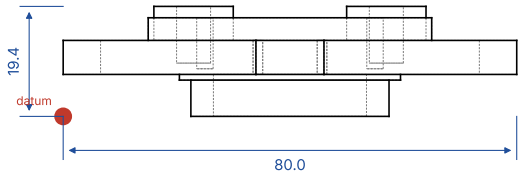
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

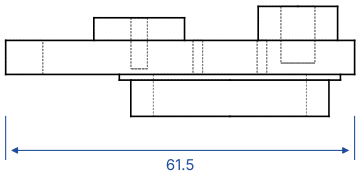


ISOMETRIC (not to scale)




FRONT VIEW




RIGHT VIEW

**HOLE TABLE (letters mark hole centres on the view along that axis)**

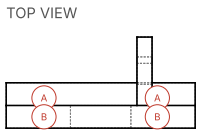
| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 27.00 | 1 | Z | 7.4 | (40.0, 39.5) | recess for the Ø25 MG996R horn |
| B | 6.00 | 2 | Z | 10.0 | (23.0, 51.5); (57.0, 51.5) | 6 mm rod press fit |
| C | 2.90 | 2 | Z | 9.0 | (25.0, 23.5); (55.0, 23.5) | #6-32 pilot (self-threading) |
| D | 1.70 | 4 | Z | 6.0 | (34.3, 33.9); (34.3, 45.2) +2 | servo-bag screw pilot |

| | |
|---|---|
| PART NO. | **CA-201** |
| TITLE | **Turret plate** |
| MATERIAL | PETG printed, 0.4 mm nozzle |
| MASS | 24.4 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 3:4 |
| UNITS | mm · general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/turret_plate.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 4 of 13 |
| DRAWN | ME 2331 Project 2 team 2026-11-02 |

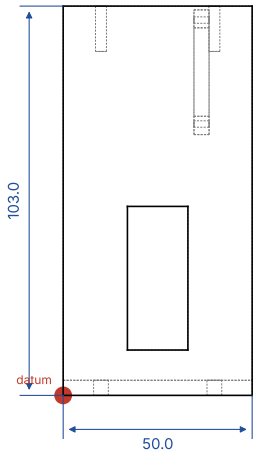
# CA-202  Spine

**NOTES**

1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.



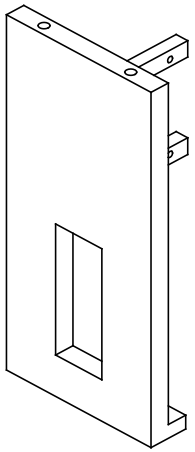
TOP VIEW



FRONT VIEW



RIGHT VIEW


ISOMETRIC (not to scale)

**HOLE TABLE  (letters mark hole centres on the view along that axis)**

| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 3.90 | 2 | Z | 4.0 | (10.0, 8.0); (40.0, 8.0) | #6-32 clearance / running |
| B | 2.90 | 2 | Z | 12.0 | (10.0, 3.0); (40.0, 3.0) | #6-32 pilot (self-threading) |
| C | 1.70 | 2 | X | 4.0 | (18.0, 71.8); (18.0, 99.3) | servo-bag screw pilot |

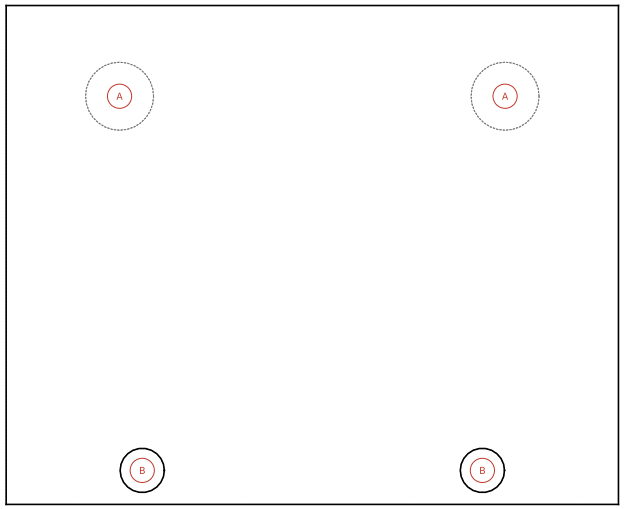
| | |
|---|---|
| PART NO. | **CA-202** |
| TITLE | **Spine** |
| MATERIAL | PETG  printed, 0.4 mm nozzle |
| MASS | 22.4 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 1:2 |
| UNITS | mm  ·  general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/spine.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 5 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |

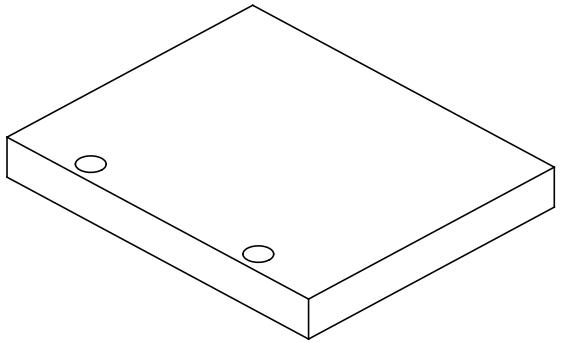
# CA-203  Top cap

**NOTES**

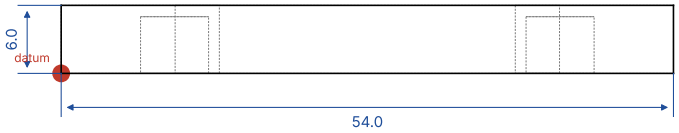
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

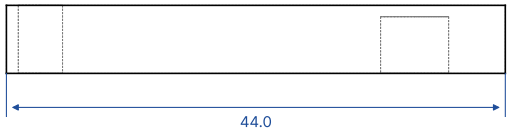


ISOMETRIC (not to scale)




FRONT VIEW




RIGHT VIEW

**HOLE TABLE  (letters mark hole centres on the view along that axis)**

| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 6.00 | 2 | Z | 5.0 | (10.0, 36.0); (44.0, 36.0) | 6 mm rod press fit |
| B | 3.90 | 2 | Z | 6.0 | (12.0, 3.0); (42.0, 3.0) | #6-32 clearance / running |

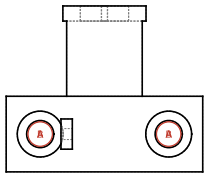
| | |
|---|---|
| PART NO. | **CA-203** |
| TITLE | **Top cap** |
| MATERIAL | PETG  printed, 0.4 mm nozzle |
| MASS | 10.5 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 3:2 |
| UNITS | mm  ·  general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/top_cap.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 6 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |

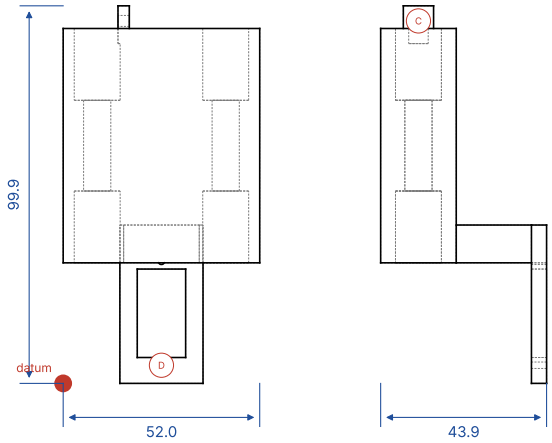
# CA-301 Carriage + arm

**NOTES**

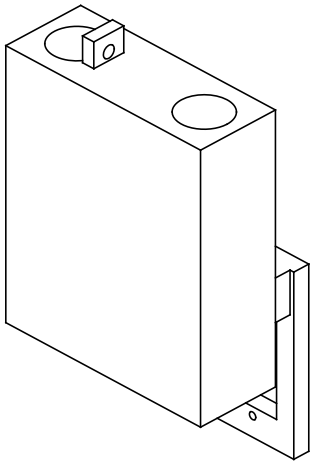
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW




FRONT VIEW RIGHT VIEW



ISOMETRIC (not to scale)

**HOLE TABLE (letters mark hole centres on the view along that axis)**

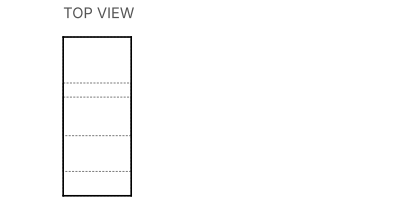
| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 12.15 | 4 | Z | 19.0 | (9.0, 10.0) ×2; (43.0, 10.0) ×2 | LM6UU press fit |
| B | 7.20 | 2 | Z | 24.0 | (9.0, 10.0); (43.0, 10.0) | #6 pan-head pocket |
| C | 2.90 | 1 | X | 3.0 | (10.0, 95.9) | #6-32 pilot (self-threading) |
| D | 1.70 | 1 | Y | 4.0 | (26.0, 4.8) | servo-bag screw pilot |

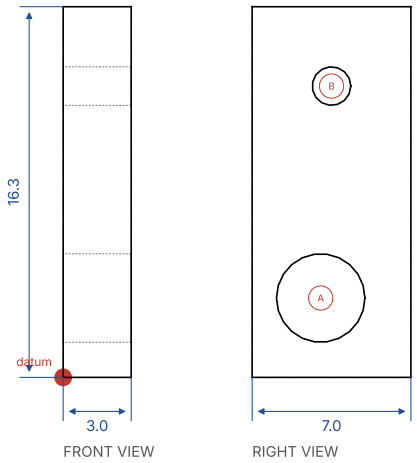
| | |
|---|---|
| PART NO. | **CA-301** |
| TITLE | **Carriage + arm** |
| MATERIAL | PETG printed, 0.4 mm nozzle |
| MASS | 46.0 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 1:2 |
| UNITS | mm · general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/carriage_arm.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 7 of 13 |
| DRAWN | ME 2331 Project 2 team 2026-11-02 |

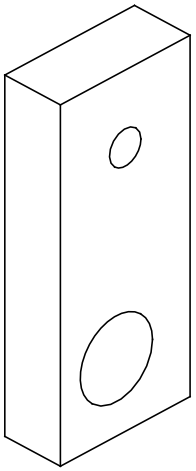
# CA-302  Lift link

**NOTES**

1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.



ISOMETRIC (not to scale)

**HOLE TABLE  (letters mark hole centres on the view along that axis)**

| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 3.90 | 1 | X | 3.0 | (3.0, 3.5) | #6-32 clearance / running |
| B | 1.70 | 1 | X | 3.0 | (3.5, 12.8) | servo-bag screw pilot |

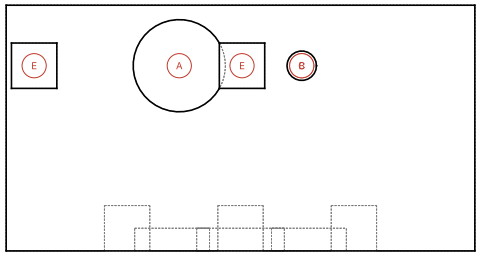
| | |
|---|---|
| PART NO. | **CA-302** |
| TITLE | **Lift link** |
| MATERIAL | PETG  printed, 0.4 mm nozzle |
| MASS | 0.2 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 3:1 |
| UNITS | mm  ·  general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/lift_link.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 8 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |

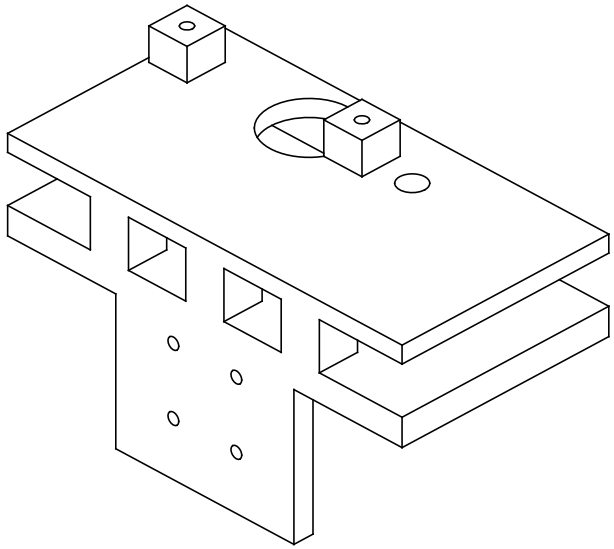
# CA-401 Gripper frame

**NOTES**

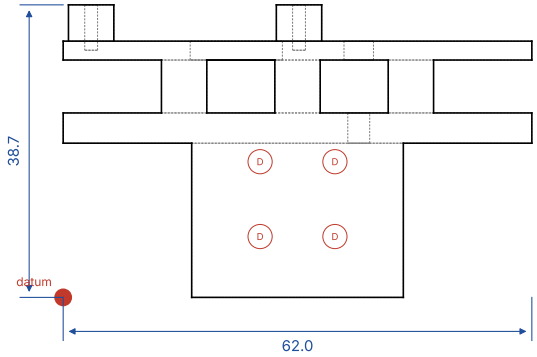
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

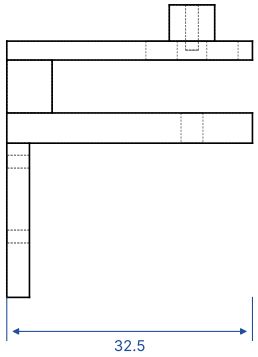


ISOMETRIC (not to scale)




FRONT VIEW




RIGHT VIEW

**HOLE TABLE (letters mark hole centres on the view along that axis)**

| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 12.20 | 1 | Z | 2.5 | (22.9, 24.5) | SG90 spline + boss clearance |
| B | 3.90 | 1 | Z | 2.5 | (39.1, 24.5) | #6-32 clearance / running |
| C | 2.90 | 1 | Z | 4.0 | (39.1, 24.5) | #6-32 pilot (self-threading) |
| D | 1.70 | 4 | Y | 3.0 | (26.1, 8.1); (26.1, 17.9) +2 | servo-bag screw pilot |
| E | 1.70 | 2 | Z | 6.0 | (3.7, 24.5); (31.2, 24.5) | servo-bag screw pilot |

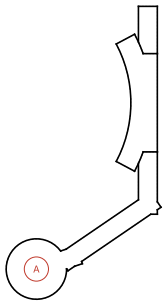
| | |
|---|---|
| PART NO. | **CA-401** |
| TITLE | **Gripper frame** |
| MATERIAL | PETG printed, 0.4 mm nozzle |
| MASS | 11.8 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 1:1 |
| UNITS | mm · general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/gripper_frame.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 9 of 13 |
| DRAWN | ME 2331 Project 2 team 2026-11-02 |

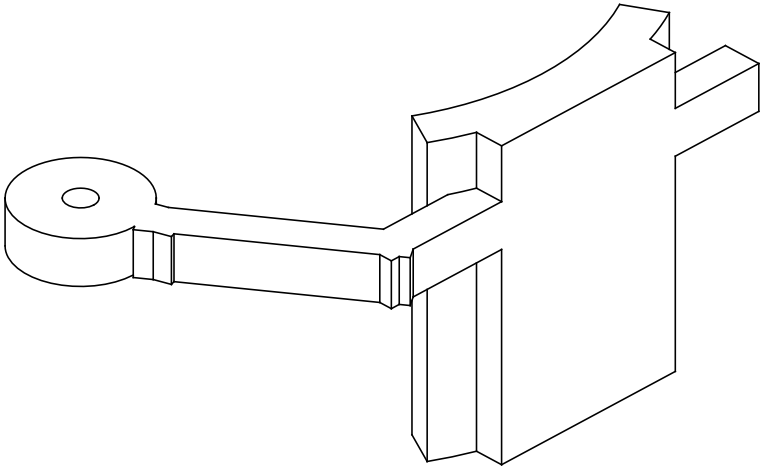
# CA-402  Finger, right

**NOTES**

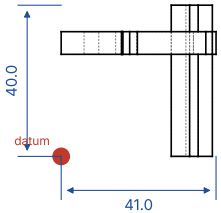
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

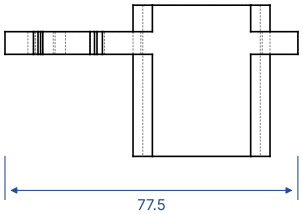


ISOMETRIC (not to scale)




FRONT VIEW




RIGHT VIEW

**HOLE TABLE  (letters mark hole centres on the view along that axis)**

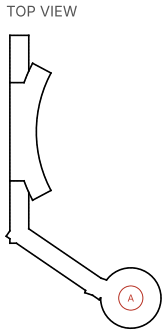
| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 3.90 | 1 | Z | 6.0 | (8.0, 8.0) | #6-32 clearance / running |

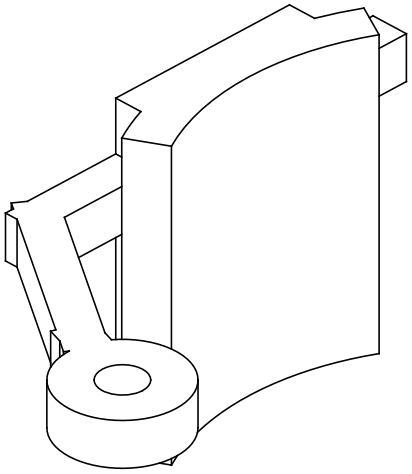
| | |
|---|---|
| PART NO. | **CA-402** |
| TITLE | **Finger, right** |
| MATERIAL | PETG  printed, 0.4 mm nozzle |
| MASS | 9.7 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 1:2 |
| UNITS | mm  ·  general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/finger_R.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 10 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |

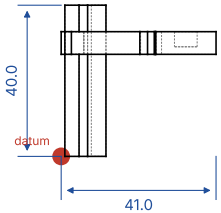
# CA-403  Finger, left

**NOTES**

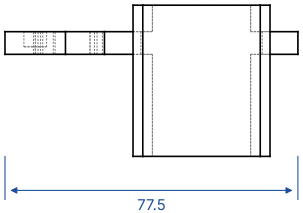
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.





ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

**HOLE TABLE  (letters mark hole centres on the view along that axis)**

| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 6.00 | 1 | Z | 4.0 | (33.0, 8.0) | SG90 spline / horn socket |

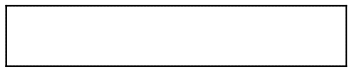
| | |
|---|---|
| PART NO. | **CA-403** |
| TITLE | **Finger, left** |
| MATERIAL | PETG  printed, 0.4 mm nozzle |
| MASS | 9.7 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 1:2 |
| UNITS | mm  ·  general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/finger_L.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 11 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |

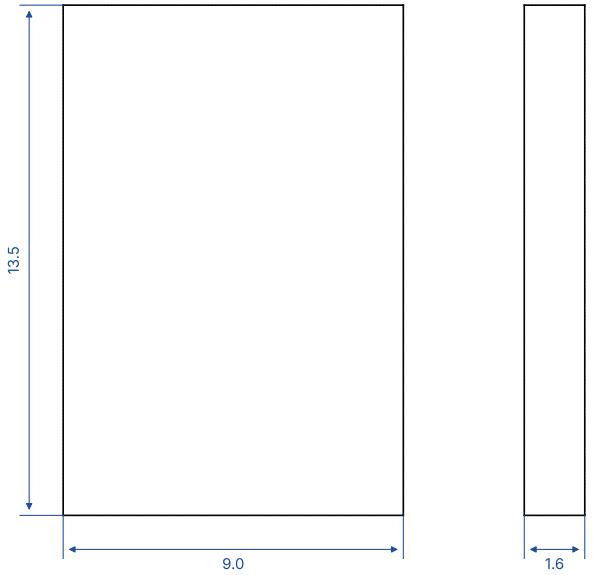
# CA-404  Can bumper

**NOTES**

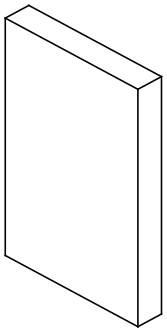
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW






FRONT VIEW  RIGHT VIEW



ISOMETRIC (not to scale)

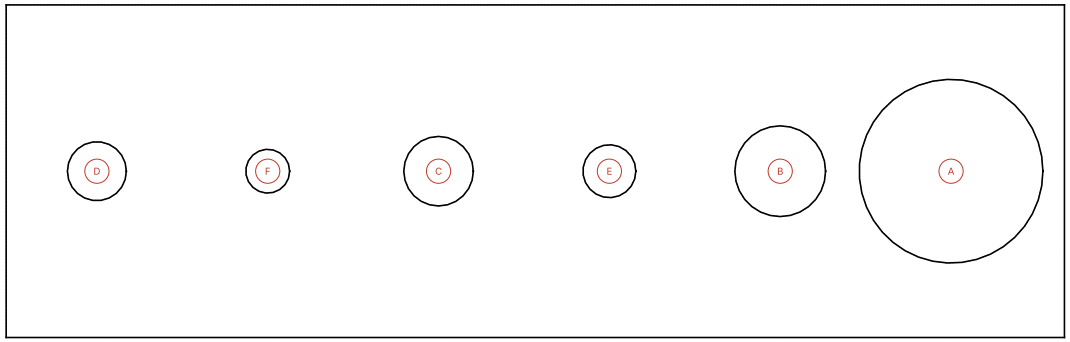
| | |
|---|---|
| PART NO. | **CA-404** |
| TITLE | **Can bumper** |
| MATERIAL | silicone  (cut from sheet) |
| MASS | 0.1 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 5:1 |
| UNITS | mm  ·  general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/can_bumper.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 12 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |

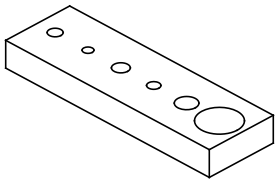
# CA-900  Tolerance coupon

**NOTES**

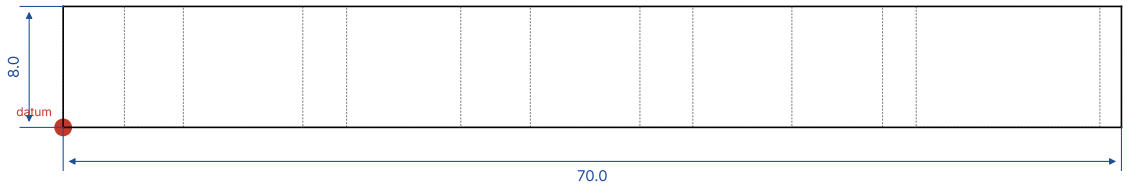
1. Hole sizes are the printed sizes from the PETG tolerance table (cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

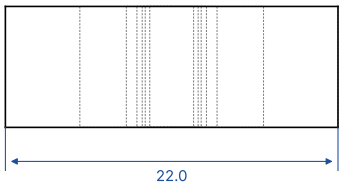


ISOMETRIC (not to scale)




FRONT VIEW




RIGHT VIEW

**HOLE TABLE  (letters mark hole centres on the view along that axis)**

| | Ø (mm) | × | axis | depth | centres from datum (mm) | fit / feature |
|---|---|---|---|---|---|---|
| A | 12.15 | 1 | Z | 8.0 | (62.5, 11.0) | LM6UU press fit |
| B | 6.00 | 1 | Z | 8.0 | (51.2, 11.0) | 6 mm rod press fit |
| C | 4.60 | 1 | Z | 8.0 | (28.6, 11.0) | #8-32 clearance |
| D | 3.90 | 1 | Z | 8.0 | (6.0, 11.0) | #6-32 clearance / running |
| E | 3.50 | 1 | Z | 8.0 | (39.9, 11.0) | #8-32 pilot |
| F | 2.90 | 1 | Z | 8.0 | (17.3, 11.0) | #6-32 pilot (self-threading) |

| | |
|---|---|
| PART NO. | **CA-900** |
| TITLE | **Tolerance coupon** |
| MATERIAL | PETG  printed, 0.4 mm nozzle |
| MASS | 8.2 g (CAD, 60 % fill) |
| QTY | 1 per robot |
| SCALE | 2:1 |
| UNITS | mm  ·  general tolerance ±0.2 |
| SOURCE | cad/model.py → cad/out/parts/tolerance_coupon.step / .stl |
| PROJECTION | Third angle ⊕◁ |
| SHEET | 13 of 13 |
| DRAWN | ME 2331 Project 2 team  2026-11-02 |